DGMS 2019 • SECTION M: SATELLITE REMOTE SENSING & PHOTOGRAMMETRY

Question 93: Satellite Image Geometry: Earth Curvature Manifestation vs. Sensor Altitude

Official Examination Guide • Geometric Distortions, Ground Swath Scales & Tangent Plane Approximations

OFFICIAL EXAMINATION QUESTION

In the satellite images, when other factors are same

(a) Equal curvature of the earth is manifested by all altitude sensors

(b) Least curvature of the earth is manifested by highest altitude sensor

(c) Least curvature of the earth is manifested by lowest altitude sensor CORRECT ANSWER

(d) Highest curvature of the earth is manifested by lowest altitude sensor

1. Executive Summary & Core Principle

In photogrammetry, geodesy, and satellite remote sensing (Lillesand, Kiefer & Chipman, *Remote Sensing and Image Interpretation*; Wolf & Dewitt, *Elements of Photogrammetry*):

- **Geometric Effect of Altitude:** The manifestation of the Earth's curvature in an optical image depends on the proportion of the Earth's spherical surface encompassed within the sensor's field of view (FOV).

- **Lowest Altitude Sensor (Option c is TRUE):** A sensor positioned at a low orbital altitude (or airborne platform) covers a relatively small ground swath (D). Because the radius of the Earth ($R \approx 6,371\text{ km}$) is massive compared to a localized low-altitude scene footprint ($D \ll R$), the vertical deviation of the spherical surface from a local tangent plane is negligible:

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  Δh ≈ D² / (2 · R)
  ```

  For a 10 km swath, $\Delta h \approx 7.8\text{ meters}$, which is practically imperceptible in the image. The ground behaves nearly as a flat Euclidean plane, and **the least curvature of the Earth is manifested**.

- **Highest Altitude Sensor:** As orbital altitude increases to thousands of kilometers (or geostationary orbit at $35,786\text{ km}$), the sensor images full continental landmasses or an entire hemisphere. The limb of the Earth forms a pronounced circular arc, and tangential distortion due to spherical curvature reaches its maximum across the image frame.

Therefore, when other factors are held equal, the **least curvature of the earth is manifested by the lowest altitude sensor**, corresponding to Option **(c)**.

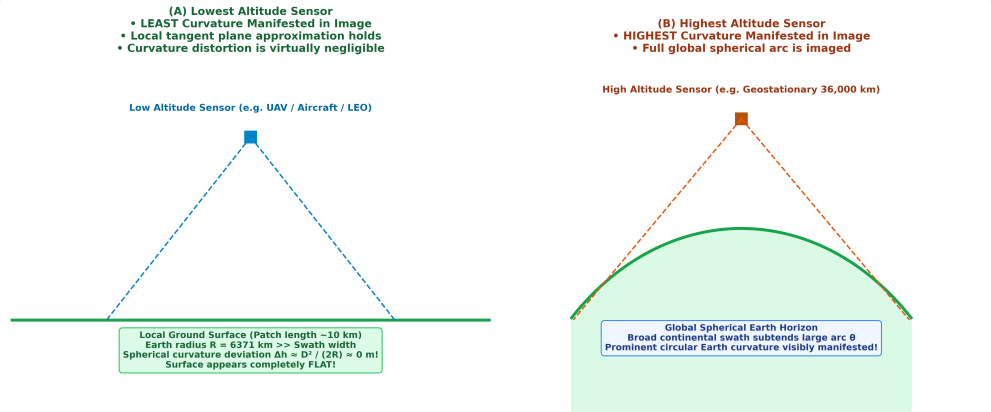


Figure 93.1: Geometric manifestation of Earth curvature comparing low-altitude sensors (planar approximation, least curvature) with high-altitude orbital sensors (global spherical arc, maximum curvature) (Grounded in Wolf & Dewitt Photogrammetry).

2. Mathematical Formulation of Earth Curvature Displacement

In aerial and satellite photogrammetry, image displacement due to Earth curvature (d_{cr}) is expressed as:

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dcr = r³ / (2 · R · H)

Where:
• r = Radial distance from the image principal point (nadir)
• R = Mean radius of the Earth (~6,371 km)
• H = Flying height / sensor altitude above ground
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Over localized scenes captured by low-altitude sensors, r is very small, meaning radial displacement from curvature is essentially zero. As altitude and angular coverage expand, planetary spherical curvature dominates the projection.

3. Evaluation of Options

- **(a) Equal curvature of the earth is manifested by all altitude sensors: INCORRECT.** Curvature manifestation is strongly altitude- and footprint-dependent.
- **(b) Least curvature of the earth is manifested by highest altitude sensor: INCORRECT.** Highest altitude sensors image the greatest curvature (the spherical limb of the Earth).
- **(c) Least curvature of the earth is manifested by lowest altitude sensor: CORRECT ANSWER.** Lowest altitude captures the smallest localized ground area where the planar tangent approximation is closest to truth.
- **(d) Highest curvature of the earth is manifested by lowest altitude sensor: INCORRECT.** Directly contradicts geometric reality.

> **Key Takeaway:** Lower altitude = smaller ground swath = closer to flat plane = **least curvature manifested**. Hence, **(c)** is the correct answer.

← Previous: Q.92 Polarization of Reflected Light | **Master Portal Index** | **Next: Q.94 Stellar Nucleosynthesis beyond Bi-209 →**